Evan J. Turner

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EDUCATION

University of Wisconsin-Madison, Madison, WI *Expected Dec 2026*
M.S., Mechanical Engineering | GPA: 3.89
Teaching Assistant: Computer-Aided Engineering (ME 331)

Clarkson University, Potsdam, NY *May 2024*
B.S. with Honors, Mechanical Engineering (Minor: Biomedical Engineering) | GPA: 3.76
Leadership: Makerspace Mentor, President & Captain - Men's Club Ice Hockey

RELEVANT EXPERIENCE

Production Engineer (Part Time), Calimetrix LLC *Aug 2026 – Present*
- Designing device housing enclosures in SolidWorks, translating prototypes into scalable injection molded products using Design for Manufacturing (DFM) principles.
- Assessing and optimizing production workflows to reduce assembly bottlenecks and scale production efficiency.

Graduate Research Assistant, Cardiovascular Fluid Dynamics Lab (UW-Madison) *Aug 2024 – Present*
- Developed an image processing and low-fidelity CFD pipeline (MATLAB/Python/SimVascular) using Doppler US to evaluate wall shear stress metrics across N = 82 pregnant subjects, presenting a method to rapidly identify patient groups with heightened risk of postpartum hypertensive disorder progression.
- Prototyped novel catheterization tools in SolidWorks using SLA/FDM 3D printing to improve procedural efficiency for interventional radiology workflows; beginning the approval process to validate the design in large animal models.
- Designed and instrumented an experimental fluidic benchtop setup to measure nephrostomy tube obstruction following exchange procedures, introducing quantitative metrics to aid the replacement decision in clinic.
- Executed patient-specific physiologic CFD simulations to predict energy loss and flow distribution for potential surgical treatments, directly guiding clinical decisions for a time-sensitive Fontan case.
- Engineered a non-invasive MRI and CFD framework to evaluate urethral biomechanics in stricture patients.

Biomedical Engineering Co-op, ZOLL Medical Corporation *Jan 2023 – Aug 2023*
- Developed a novel polymer fabrication process for a concept sensor, increasing process yield from <5% to ~100%, boosting throughput by ~40x, and enhancing device sensitivity by 10x.
- Designed and fabricated test fixtures in SolidWorks using 3D printing and laser cutting to mechanically verify sensor performance.
- Wrote and executed engineering design studies to validate new product performance using custom-built MATLAB signal processing tools for time-frequency data analysis.

Research Assistant, Connective Tissue Mechanics Lab (Clarkson University) *May 2021 – May 2024*
- Developed an ex-vivo model to quantify post-retraction leakage in intradiscal injections, establishing a benchmark for rapid risk screening. Led a study using this technique to investigate post-procedure movement as a risk factor.
- Designed a low-cost, wearable spinal motion tracking system utilized across multiple biomechanics studies, resulting in a publication on spinal motion coordination post-fatigue in competitive Nordic skiers.

PUBLICATIONS & PRESENTATIONS

- **Journal Article:** "Establishing a Through-Puncture Model for Assessing Post-Injection Leakage", European Spine Journal, 2022.
- **Conference Presentation: "**WSS-Based Differentiation of Hypertensive Disorders of Pregnancy Using US-Based CFD", ASME SBC, 2025. *(Finalist, MS Student Paper Competition)*
- **Journal Article: "**Biomechanical Analysis of Lumbar Coordination in Nordic Skiers During Double Poling", Communications in Kinesiology, 2026.
- **Journal Article:** "Non-Invasive Methods for Bladder Biomechanics During Voiding Using Uro-Dynamic MRI", Bioengineering, 2026.
- **Conference Presentation:** "MRI-Based CFD in Urethral Stricture Biomechanics", ISMRM Annual Meeting, 2025.

TECHNICAL SKILLS

- **CAD & Mechanical Design:** SolidWorks (CSWA), AutoCAD, Fusion360, KiCAD, DFM, GD&T
- **Programming & Simulation:** MATLAB, Python, C++, LabVIEW, ANSYS, Altair, SimVascular
- **Manufacturing & Fabrication:** 3D Printing (FDM/SLA), Laser Cutting, CNC Routing, Lathe, Soldering